

Individual Pathways

Activity 6.1

Revising the atom

1. Identify the set that contains an element, a compound and a mixture.
 - (a) carbon dioxide, air, silver
 - (b) gold, sulfur, phosphorus
 - (c) granite, oxygen, hydrogen
 - (d) copper oxide, calcium carbonate, sodium chloride
2. Identify the correct statement.
 - (a) Niels Bohr demonstrated that atoms combine in whole number ratios.
 - (b) Democritus discovered the atomic nucleus.
 - (c) J.J. Thomson discovered the electron.
 - (d) Rutherford discovered the neutron.
3. Select the correct statement about the atom.
 - (a) Protons have the same electrical charge as neutrons.
 - (b) Rutherford used alpha particles from a radioactive element to investigate the sub-structure of the atom.
 - (c) The hydrogen-1 isotope consists of one proton, one neutron and one electron.
 - (d) All the electrons in an atom are located at the edge of the spherical atom.
4. Select the correct statement about hydrogen isotopes.
 - (a) H-2 contains two protons, two neutrons and one electron
 - (b) H-1 contains one neutron and one electron
 - (c) H-1 contains one proton and one neutron
 - (d) H-3 contains one proton, two neutrons and one electron
5. Select the statement that is true of carbon-14.
 - (a) Its nucleus contains six neutrons.
 - (b) It has one more neutron than carbon-12.
 - (c) It has two more electrons than carbon-12.
 - (d) It is radioactive.

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6. Beta particles are
 - (a) fast-moving electrons.
 - (b) positively charged.
 - (c) heavier than an alpha particle.
 - (d) unable to penetrate human skin.

7. The radioactive element present in smoke detectors is
 - (a) americium-241.
 - (b) uranium-238.
 - (c) strontium-90.
 - (d) carbon-14.

8. Gamma rays consist of
 - (a) streams of neutrons.
 - (b) high energy electromagnetic radiation.
 - (c) positive protons.
 - (d) negative particles travelling at 85% of the speed of light.

9. In nuclear symbolism, the letter 'Z' stands for
 - (a) the atomic number — the number of neutrons in the nucleus.
 - (b) the atomic number — the number of protons in the nucleus.
 - (c) the number of neutrons.
 - (d) the mass number.

10. Consider the nuclear symbol ${}_{92}^{235}\text{U}$. Select the true statement relating to this symbol.
 - (a) Uranium atoms contain more protons than neutrons.
 - (b) There are 235 electrons orbiting the nucleus.
 - (c) The atomic number of uranium is 235.
 - (d) The nucleus contains 143 neutrons.

11. Cosmic radiation mainly consists of
 - (a) X-rays.
 - (b) infra-red rays.
 - (c) high energy protons.
 - (d) electrons.

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12. Iodine -131 has a half-life of 8 days. If a sample of I-131 has a mass of 60 g, how much radioactive iodine will be left after 24 days?
- (a) 7.5 g
 - (b) 15 g
 - (c) 30 g
 - (d) 3.75 g
13. The symbol for a gamma ray is
- (a) α
 - (b) λ
 - (c) β
 - (d) γ
14. Answer true or false for each statement.
- (a) Ammonia is a compound.
 - (b) Calcium is an element.
 - (c) John Dalton proposed that matter contained positive and negative charges.
 - (d) Isotopes have different mass numbers.
 - (e) H-3 is a radioisotope.
15. Answer true or false for each statement.
- (a) Carbon-12 has six neutrons.
 - (b) Alpha particles contain two protons and two neutrons.
 - (c) Rutherford's atomic model had electrons orbiting the nucleus.
 - (d) Bohr's atomic model had electrons and neutrons orbiting the nucleus.
 - (e) Neutrons have a negative charge.
16. Answer true or false for each statement.
- (a) Radioactive elements are often stored in lead boxes.
 - (b) Alpha particles pose a great external danger to the body.
 - (c) There are about 50 natural radioisotopes.
 - (d) Beta particles travel about 100 cm through the air.
 - (e) Becquerel accidentally discovered radioactivity while investigating uranium compounds.

Answers

1. a
2. c
3. b
4. d
5. d
6. a
7. a
8. b
9. b
10. d
11. c
12. a
13. d
14. (a) True (b) True (c) False (d) True (e) True
15. (a) True (b) True (c) True (d) False (e) False
16. (a) True (b) False (c) True (d) True (e) True